Lecture 9

Readings beyond spectral measures: POVMs

Objectives. *A pointer can have noisy statistics that are not a spectral measure of the system alone. Effects describe those probabilities. We prove that every countable family of such effects can be obtained by a sharp reading on a larger system, derive the effects of a position pointer, and identify the obstruction to a joint sharp position–momentum reading.*

- Define a POVM and prove its countable Naimark dilation, including a valid enlarged unitary implementation.
- Compute the calibrated pointer's effect, keeping the reflection of its response density.
- Prove the joint-sharp-reading commutation criterion and evaluate the coherent-state phase-space reading.

9.1 Probabilities need effects, not necessarily projections

QM-I's probe (Example 11.11) already formed $F_j = K_j^\dagger K_j$ and called its family a POVM. These are the effects defined generally in Lecture 3. No requirement made $F_j^2 = F_j$. For a whole reading procedure, its events must fit together into one normalised probability assignment for every state.

For elementary real-outcome calculations, events are intervals and their finite disjoint unions. In phase space use rectangles and finite disjoint unions of rectangles. An additivity statement assumes its union belongs to this event class. The general cited theorems below use *Borel sets*, the larger class generated by countable unions and complements; their full event-space framework is a cited input, not a measure-extension construction supplied here.

Definition 9.1 (Positive operator-valued measure)**.** For countably many outcomes, a *POVM* is a family of effects $F_j \geq 0$ with $\sum_j F_j = \mathbb{1}$ strongly. Its outcome probabilities are $p_j = \mathrm{Tr}(\rho F_j)$. For real or phase-space outcomes it assigns effects $F(B)$ to event sets, with $F(\varnothing) = 0$, $F(\text{whole space}) = \mathbb{1}$, and, for every disjoint countable event family,

$$\left\langle \psi, F\Big(\bigcup_j B_j\Big)\psi \right\rangle = \sum_j \left\langle \psi, F(B_j)\psi \right\rangle .$$

A POVM is *sharp* when its effects form a projection-valued measure (PVM), with $F(B\cap C) = F(B)F(C)$; otherwise it is unsharp.

This is QM-II's interval spectral-measure structure with multiplicativity dropped. Finite additivity extends to finite unions, and countable additivity is required whenever the union

is still an admissible event.

Positivity gives $0 \le F(B) \le \mathbb{1}$ by its complement. The countable axiom is deliberately weaker than multiplicativity. L1's spectral density expansion and interchange of nonnegative sums show it gives countably additive scalar probabilities for every density operator, not just rays. A joint phase-space POVM has marginals $F_1(I) = F(I \times \mathbb{R})$ and $F_2(J) = F(\mathbb{R} \times J)$.

The mean operator alone loses information. For $0 < \eta < 1$, a noisy qubit reading with numerical outputs ± 1 has effects $F_\pm = (\mathbb{1} \pm \eta\sigma_z)/2$ and first moment $F_+ - F_- = \eta\sigma_z$. The sharp operator $A = \eta\sigma_z$ has that same mean operator, but its outputs are $\pm\eta$ with spectral projections $(\mathbb{1} \pm \sigma_z)/2$. In $|0\rangle$ the noisy reading gives ± 1 with probabilities $(1 \pm \eta)/2$, while the sharp one gives $+\eta$ with certainty. Their second moment operators are $\mathbb{1}$ and $\eta^2\mathbb{1}$ respectively. Thus specifying a first moment does not specify the reading's statistics. The identification of an observable with one self-adjoint operator includes the additional assertion that its statistics are sharp.

Self-check 9.1

1. For $\eta = 1/2$, list the two reading distributions in $|0\rangle$ and check their equal mean.
2. Which PVM axiom is absent from a general POVM?

9.2 Naimark: a sharp reading on a larger space

Theorem 9.2 (Countable Naimark dilation). *For a countable POVM on nonzero separable $\mathcal{H}$, pad a finite list with zero effects if necessary and let (e_j) be the basis of ℓ^2. There is an isometry $J : \mathcal{H} \to \mathcal{H} \otimes \ell^2$ with*

$$J\psi = \sum_j \sqrt{F_j}\psi \otimes e_j, \qquad F_j = J^\dagger(\mathbb{1} \otimes |e_j\rangle\langle e_j|)J.$$

With an additional two-dimensional factor this is implemented by a unitary and fixed ready ancilla, with $U_M(\psi \otimes e_0 \otimes f_0) = J\psi \otimes f_0$ and pointer $\Pi_j = \mathbb{1} \otimes |e_j\rangle\langle e_j| \otimes \mathbb{1}_2$. The same probabilities are a sharp pointer reading there.

Proof. S6 supplies each positive square root. Orthogonality of the register coordinates gives

$$\|J\psi\|^2 = \sum_j \left\|\sqrt{F_j}\psi\right\|^2 = \sum_j \langle \psi, F_j\psi\rangle = \|\psi\|^2 .$$

The series therefore converges in the L6 tensor norm. It is linear by limits of finite sums; polarization gives $J^\dagger J = \mathbb{1}$. Pairing arbitrary φ, ψ with the jth register projection gives $\langle\sqrt{F_j}\varphi, \sqrt{F_j}\psi\rangle = \langle\varphi, F_j\psi\rangle$, proving the compressed-effect formula.

Write $\mathcal{K}_0 = \mathcal{H} \otimes \ell^2$ and $\mathcal{K} = \mathcal{K}_0 \otimes \mathbb{C}^2$, with last-factor basis f_0, f_1. Define an isometry between the input subspace and image by

$$\psi \otimes e_0 \otimes f_0 \longmapsto J\psi \otimes f_0.$$

Both subspaces are closed: the input has its explicit product projection, and the image projection is $J'J'^\dagger$, where $J'\psi = J\psi \otimes f_0$. Both orthogonal complements contain $\mathcal{K}_0 \otimes f_1$, so both have countably infinite Hilbert dimension. Complement ONBs can be constructed by applying the complementary projections to a countable basis of $\mathcal{K}$ and using Gram–Schmidt on their dense spans. Map these ONBs onto one another, extending the displayed isometry to an onto unitary U_M. With ready ancilla $e_0 \otimes f_0$, read the pointer projections

$\Pi_j = \mathbb{1} \otimes |e_j\rangle\langle e_j| \otimes \mathbb{1}_2$. Their probabilities are exactly $\mathrm{Tr}(\rho F_j)$, first on rays by the compression and then on density operators by their positive spectral sums. ∎

The spare factor makes both complements visibly infinite without computing the complement of $J(\mathcal{H})$, even for infinite outcome lists. An arbitrary infinite isometry need not extend on the original space without it. This construction proves existence of a reading model; it does not specify a unique apparatus or post-reading state. L10 will distinguish procedures with the same POVM.

For real Borel outcomes the general Naimark theorem is *cited*: every normalised POVM is a compression $F(B) = J^\dagger E(B) J$ of a PVM on a larger space (Busch–Lahti–Pellonpää–Ylinen, abbreviated BLPY below, Theorem 7.7 p. 146). Its proof is not the replacement of the countable sum above by an unexplained continuous basis of eigenvectors.

Self-check 9.2

1. Write J explicitly for the noisy-qubit effects $F_\pm$ in §9.1, using two register coordinates.
2. What ensures that the infinite-dimensional isometry extends to an onto unitary here?

9.3 A calibrated position pointer

Take system coordinate x and meter coordinate y, a normalised $\varphi \in \mathcal{S}(\mathbb{R})$ and coupling strength $g > 0$. Define the coupling first by the shear

$$(U_g \Psi)(x, y) = \Psi(x, y - gx).$$

A change of y variable proves norm preservation for smooth compactly supported functions; S4 extends it to an isometry of $L^2(\mathbb{R}^2)$, and U_{-g} is its inverse. In the meter's partial Fourier representation it is multiplication by $e^{-igxp_M/\hbar}$. The notation $e^{-igQ\otimes P^{(2)}/\hbar}$ denotes this unitary; the shear definition keeps us from assuming a domain for a formal unbounded operator product. For a Schwartz system input it gives $\psi(x)\varphi(y - gx)$.

Calibrate the meter reading as $\hat{x} = y/g$. The probability that it lies in I is

$$\begin{aligned} p_\psi(I) &= \int |\psi(x)|^2 \int_{gI} |\varphi(y - gx)|^2 \, dy \, dx \\ &= \int |\psi(x)|^2 f_I(x) \, dx, \qquad f_I(x) = \int_I g|\varphi(g(z - x))|^2 \, dz. \end{aligned}$$

All integrands are nonnegative and the total is one. Thus $0 \le f_I \le 1$, and $F(I) = f_I(Q)$ is an effect. The interval effect also has the bounded compression form

$$F(I) = J_\varphi^\dagger U_g^\dagger (\mathbb{1} \otimes E_{Q^{(2)}}(gI)) U_g J_\varphi, \qquad J_\varphi \psi = \psi \otimes \varphi.$$

The known meter interval PVM is strongly additive; bounded compression therefore proves additivity for admissible unions and finite unions. Both sides of the probability identity are continuous quadratic forms in ψ, so S4 extends the Schwartz calculation to every L^2 vector. Positive spectral sums then give the same effect for every density operator. No position eigenvector was an input state.

With convolution defined as $(f * e)(x) = \int f(z) e(x - z) \, dz$, the result is

$$F(I) = (1_I * e)(Q), \qquad e(u) = g|\varphi(-gu)|^2. \tag{9.1}$$

The reflection is essential. The response's *actual reading error* $\hat{x} - x$ has density $h(u) = g|\varphi(gu)|^2 = e(-u)$; e is the kernel used in the effect convolution, not that error density. A

meter initially centred at μ shifts the readout by $+\mu/g$, although e is centred at $-\mu/g$. Known bias can be removed in calibration. For a centred even Gaussian the two densities coincide, which can conceal a wrong sign.

The same response density has two roles in this formula. For a fixed input coordinate x, the density of the calibrated output z is $h(z-x)$. The interval probability at that coordinate is therefore

$$f_I(x) = \int_I h(z-x)\,dz.$$

The convolution convention uses the argument $x-z$ instead. It consequently requires $e(x-z) = h(z-x)$. Figure 9.1 shows why a displaced response makes the sign visible. Here x is an integration coordinate, not a proposed exactly sharp preparation.

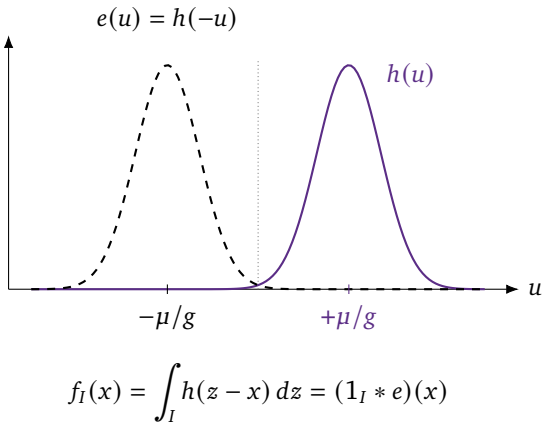


Figure 9.1: The reading-error density h (solid) and the reflected effect kernel e (dashed) for a Gaussian meter with positive bias. The widths agree, but their centres have opposite signs; the effect integrates the output density over the chosen interval.

For example, with a Gaussian meter and a bounded interval of positive length, both its interval and complement integrals are strictly positive. Thus $0 < f_I(x) < 1$ for every x. An interval effect is generally not a projection: where $0 < f_I(x) < 1$, its square multiplies by $f_I(x)^2 \neq f_I(x)$. Finite meter width produces blurred edges instead of a sharp interval event. This implements QM-II's promised unsharp position reading with a concrete response function. For multiplication operators, equality *almost everywhere* means equality outside a set of positions of length zero. Changing values on such a set does not change the represented L^2 function or multiplication operator. Later we will compute its state change and the momentum spread added by ignoring the meter; neither is specified by (9.1) alone.

Self-check 9.3

1. The meter wavefunction φ has mean position $\mu > 0$. Which way does the calibrated reading shift? Which way is e centred?
2. What pointwise condition makes a multiplication effect a projection?

9.4 Joint sharp readings must commute

Lemma 9.3 (An effect dominated by a projection). *If $0 \le F \le E$ and E is a projection, then $F = EFE$, so $EF = FE = F$.*

Proof. For $v \in \ker E$, $0 \le \langle v, Fv\rangle \le \langle v, Ev\rangle = 0$. S6 implies $\left\|\sqrt{F}v\right\|^2 = 0$, hence $Fv = 0$. Self-adjointness kills all cross terms with $\ker E$ as well, giving $F = EFE$. ∎

Proposition 9.4 (Joint sharp readings)**.** *Two countable PVMs have a joint POVM exactly when all their projections commute. If only the first marginal is sharp, the second marginal's effects must commute with it.*

Proof. Suppose a countable joint POVM (M_{jk}) has sharp first marginal $E_j = \sum_k M_{jk}$ and second marginal $F_k = \sum_j M_{jk}$. Since $M_{jk} \le E_j$, the lemma gives $E_j M_{jk} = M_{jk}$. For a different first label ℓ, $E_j M_{\ell k} = 0$, because $M_{\ell k}$ is supported in the orthogonal range of E_ℓ. Taking the strong marginal sum therefore yields

$$E_j F_k = M_{jk} = F_k E_j.$$

Thus a joint reading with one sharp marginal already forces the other marginal's effects to commute with it. In particular, two countable PVMs can have a joint POVM only if every pair of their projections commutes.

Conversely, for commuting PVMs (E_j), (F_k) set $M_{jk} = E_j F_k$. These are projections, mutually orthogonal for different pairs. For every vector, nonnegative sums give $\sum_{jk} \langle \psi, M_{jk}\psi\rangle = \sum_j \left\|E_j\psi\right\|^2 = \|\psi\|^2$, using completeness of the second PVM within each first-sector vector. Thus their sum is strongly $\mathbb{1}$ and their marginals are the prescribed PVMs. This proves the countable joint-sharp-reading equivalence. ∎

For real outcomes, a joint POVM M with sharp marginals E_1, E_2 gives the same necessity using the four events $I \times J$, $I \times J^c$, $I^c \times J$, $I^c \times J^c$ and the lemma; complements of intervals are finite unions of intervals, so these events remain in the elementary class. It yields $E_1(I)E_2(J) = M(I \times J) = E_2(J)E_1(I)$ for every I, J. The general continuous converse uses a joint spectral construction and is *cited*, BLPY, Theorem 11.3 pp. 269–270, for its real Borel value spaces.

For Q, P the relevant noncommutation is spectral, not merely a formal commutator on a test domain. Here is the spectral argument, *sketched* using QM-II's cited spectral calculus. If every position projection commuted with every momentum projection, it would commute with $U(a) = e^{-iaP/\hbar}$. But this unitary translates a wave packet from an interval I to $I + a$. Choose a nonzero smooth packet supported in I and a translation with disjoint intervals: projecting before or after translation gives different results. Thus some spectral projections do not commute. There is no joint POVM with both exact Q and exact P as marginals. Joint unsharp readings remain possible.

Self-check 9.4

1. Where does $F \le E$ force the cross blocks of F to vanish?
2. Does the necessity proof require both marginals to be sharp?

9.5 A coherent-state phase-space reading

Recall QM-II's coherent vectors ψ_α with oscillator length s, $\alpha = (q/s + isp/\hbar)/\sqrt{2}$. The vacuum-generated phase-space POVM has density

$$F(d^2\alpha) = \frac{1}{\pi} |\psi_\alpha\rangle\langle\psi_\alpha|\, d^2\alpha. \tag{9.2}$$

Its normalisation and marginals are *cited*, BLPY, equation (8.14) p. 174 and Proposition 8.9 p. 175. Its position marginal is position blurred by a Gaussian of variance $s^2/2$; its momentum marginal is momentum blurred by one of variance $\hbar^2/(2s^2)$. The coordinates satisfy $dq\,dp = 2\hbar\,d^2\alpha$, so this is equivalently $|\psi_\alpha\rangle\langle\psi_\alpha|\,dq\,dp/(2\pi\hbar)$.

For coherent input ψ_β, the known number-basis expansion gives

$$\langle \psi_\alpha, \psi_\beta \rangle = e^{-(|\alpha|^2+|\beta|^2)/2} \sum_{n=0}^{\infty} \frac{(\bar{\alpha}\beta)^n}{n!} = e^{-(|\alpha|^2+|\beta|^2)/2+\bar{\alpha}\beta}.$$

The reading density is therefore

$$p_\beta(\alpha) = \frac{1}{\pi} e^{-|\alpha-\beta|^2}. \tag{9.3}$$

The two real components have variance $1/2$, and the density integrates to one by the ordinary Gaussian integral. In physical coordinates the observed variances are s^2 and $\hbar^2/s^2$: each contains the coherent input's vacuum variance *plus* the reading's vacuum blur. The added standard deviations have product $\hbar/2$, whereas the total observed standard deviations have product $\hbar$. There is no exact joint reading hidden in a notation that assigns both numbers to one outcome. It is a specified noisy joint POVM.

The physics

Ideal eight-port homodyne detection, with its parameter input in the vacuum and the two readout quadratures set 90° apart, implements this phase-space reading in the strong-local-oscillator limit (BLPY, §19.4, Proposition 19.2 pp. 443–445). The parameter-port vacuum supplies the additional reading blur; another input there changes the POVM. Finite local-oscillator amplitude and inefficient detectors require further qualifications or smearing. The Gaussian above is the prediction for the ideal coherent input and ideal POVM, not a universal detector response. The extra vacuum contribution belongs to the reading as well as to the prepared state; counting the vacuum only in the state, or only in the reading, gives half of each observed variance.

Self-check 9.5

1. For vacuum input $\beta = 0$, find the density at $\alpha = 0$ and the observed position variance.
2. Why does reporting two real outcomes not contradict the sharp commutation theorem?

What does a pointer read?

A reading's statistics are a POVM. Every countable POVM is a projective reading of the system with an ancilla; real-outcome dilation is cited. Sharp statistics are the spectral measures of self-adjoint operators. Joint sharp marginals must commute, so exact Q and P cannot both be marginals of one reading. A calibrated pointer and the specified eight-port detection instead supply explicit unsharp effects. A POVM specifies probabilities; the state after a recorded outcome needs an instrument.

Summary. Effects extend sharp spectral readings. Square roots yield countable Naimark dilation, with an enlarged space ensuring unitary completion. The shear pointer gives a reflected convolution kernel. Projection domination proves the sharp joint-reading constraint. The coherent-state POVM gives a Gaussian phase-space output, with separate preparation and reading contributions to its variance.

Exercises. Try Sheet 9 in the separate *Exercise sheets* booklet (English).

Self-check answers.

9.1 1. Noisy: +1 with 3/4, −1 with 1/4; sharp: +1/2 certainly; both means 1/2. 2. Multiplicativity on event intersections.

9.2 1. $J\psi = \sqrt{F_+}\psi \otimes e_0 + \sqrt{F_-}\psi \otimes e_1$, with diagonal square roots of $(\mathbb{1} \pm \eta\sigma_z)/2$; all other coordinates vanish. 2. Both complements have countably infinite ONBs, due to the padded register and spare factor.

9.3 1. Reading bias $+\mu/g$; effect kernel centre $-\mu/g$. 2. The multiplier is zero or one almost everywhere.

9.4 1. $\sqrt{F}$ kills $\ker E$ and self-adjointness kills its cross matrix elements. 2. One sharp marginal already forces the other's effects to commute with it; the converse given concerns two sharp PVMs.

9.5 1. $1/\pi$ and s^2. 2. Its two marginals are blurred, not sharp.

Sources. Busch–Lahti–Pellonpää–Ylinen, *Quantum Measurement*, pp. 71, 146, 168–175, 269–270, 443–447; Busch–Lahti–Mittelstaedt, *The Quantum Theory of Measurement*, §III.2.6 pp. 41–43; Heinosaari–Ziman, Proposition 3.25 pp. 123–124. QM-II's cited spectral theorem is the input to the explicitly sketched Q, P argument; coherent vectors are its Theorem 8.10.